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United States Patent	12401115
Kind Code	B2
Date of Patent	August 26, 2025
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Vehicle upper structure

Abstract

A vehicle upper structure disclosed in the present specification comprises a roof panel having a communication opening formed therein; a roof reinforcement disposed near a front end of the communication opening and joined to the roof panel; a rear header disposed near a rear end of the communication opening and joined to the roof panel; and an antenna plate that supports a communication device and is connected to the roof reinforcement and the rear header, wherein the rear header is joined to the roof panel at a front end and a rear end thereof, and forms a first closed space elongated in a vehicle width direction between the rear header and the roof panel.

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Appl. No.: 18/525849

Filed: December 01, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240186688 A1	Jun. 06, 2024

Foreign Application Priority Data

JP	2022-194043	Dec. 05, 2022
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Publication Classification

Int. Cl.: H01Q1/32 (20060101); B62D25/06 (20060101); B60R11/02 (20060101)

U.S. Cl.:

CPC H01Q1/3275 (20130101); B62D25/06 (20130101); B60R11/02 (20130101)

Field of Classification Search

CPC: H01Q (1/32); H01Q (1/243); H01Q (1/3275); H01Q (1/42); H01Q (9/0407)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7847744	12/2009	Mierzwa	343/711	H01Q 1/288
11233318	12/2021	Imamura	N/A	H01Q 1/48
11456525	12/2021	Katsura	N/A	H01Q 1/3208
12057625	12/2023	Ishida	N/A	H01Q 1/3275
2021/0057806	12/2020	Katsura et al.	N/A	N/A
2022/0177040	12/2021	Asai et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2468544	12/2011	EP	N/A
2011-078020	12/2010	JP	N/A
2012131338	12/2011	JP	N/A
2022089619	12/2021	JP	N/A
2019/150666	12/2018	WO	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-194043 filed on Dec. 5, 2022, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.

TECHNICAL FIELD

(2) This specification discloses an upper structure of a vehicle.

BACKGROUND

(3) Conventionally, it has been proposed to arrange a communication device such as an antenna in an upper portion of a vehicle. For example, Patent Document 1 discloses a technology in which an opening is provided in a roof panel of a vehicle and an antenna module is disposed in the opening. In Patent Document 1, the antenna module is covered with a redome made of a resin capable of transmitting radio waves.

(4) Recently, communication devices have increased in size in order to improve communication

performance. When a large-sized communication device is disposed, the opening of the roof panel must be increased. However, if the opening is increased, the rigidity of the vehicle body at the periphery of the opening may be reduced, and deformation of the vehicle body may occur.

(5) Therefore, it has been proposed in part to form a skeleton-shaped portion around the opening. However, when an excessively large skeleton-shaped portion is provided, the communication device or the skeleton-shaped portion extends to the vehicle compartment side, and the space in the vehicle compartment becomes narrow.

(6) Accordingly, the present specification discloses a vehicle upper structure capable of achieving both the rigidity of the vehicle body and the space in the vehicle compartment.

CITATION LIST

(7) PATENT DOCUMENT 1: WO 2019/150666 A1

SUMMARY

(8) The vehicle upper structure disclosed in the present specification comprises a roof panel having a communication opening formed therein; a roof reinforcement disposed near a front end of the communication opening and joined to the roof panel; a rear header disposed near a rear end of the communication opening and joined to the roof panel; and an antenna plate that supports a communication device and is connected to the roof reinforcement and the rear header, wherein the rear header is joined to the roof panel at a front end and a rear end thereof, and forms a first closed space elongated in a vehicle width direction between the rear header and the roof panel.

(9) By joining the rear header and the roof panel and forming the first closed space between the rear header and the roof panel, high rigidity can be ensured while suppressing the dimension of the rear header in the front-rear direction. By reducing the dimension of the rear header in the front-rear direction, the communication device can be disposed rearward of the vehicle. Thus, the communication device can be prevented from protruding to the vehicle compartment side. As a result, according to the above configuration, it is possible to achieve both the rigidity of the vehicle body and the space of the vehicle compartment.

(10) In this case, the rear header may comprise a front wall portion; a vertical wall portion extending from a rear end of the front wall portion toward a lower side of the vehicle; a rear wall portion extending from a lower end of the vertical wall portion toward a rear side of the vehicle; and a groove-shaped portion formed in the rear wall portion and forming the first closed space between the groove-shaped portion and the roof panel, and wherein the rear header may be joined to the roof panel at a first joining point located at a rear side of the vehicle relative to the groove-shaped portion, a second joining point located at the vertical wall portion, and a third joining point located at the front wall portion.

(11) The rear header may form a second closed space between the rear header and the roof panel at a portion between the second joining point and the third joining point.

(12) With this configuration, the rigidity of the vehicle body in the vicinity of the rear end of the communication opening can be further improved.

(13) The antenna plate may include a mounting portion to which the communication device is mounted, a hanging portion extending from a rear end of the mounting portion toward a lower side of the vehicle, and a terminal portion extending from a lower end of the hanging portion toward a rear side of the vehicle; and an terminal portion of the antenna plate is connected to the rear wall portion of the rear header.

(14) With such a configuration, it is possible to secure a large vertical dimension of the antenna plate without narrowing the cabin. This increases the section modulus of the antenna plate, thereby further improving the rigidity of the vehicle body in the vicinity of the rear end of the communication opening.

(15) A density of a joining point between the rear header and the roof panel at an end region in the vehicle width direction of the roof panel may be higher than the density of the joining point at a center region in the vehicle width direction of the roof panel.

(16) With such a configuration, since the rigidity of the vehicle body at the vehicle width direction end portion can be increased, it is possible to effectively suppress matchbox deformation of the vehicle.

(17) According to the vehicle upper structure disclosed in the present specification, it is possible to achieve both the rigidity of the vehicle body and the space of the cabin.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Embodiment(s) of the present disclosure will be described based on the following figures, wherein:

(2) FIG. 1 is a perspective view of an upper portion of a vehicle;

(3) FIG. 2 is a plan view of the periphery of a communication device.

(4) FIG. 3 is a cross-sectional view taken along the line A-A in FIG. 2;

(5) FIG. 4 is a cross-sectional view showing an upper structure of a vehicle according to a reference example; and

(6) FIG. 5 is a diagram illustrating a difference in head clearance depending on a position of a bending point.

DESCRIPTION OF EMBODIMENT

(7) Hereinafter, an upper structure of a vehicle will be described with reference to the drawings.

FIG. 1 is a perspective view of an upper portion of a vehicle; FIG. 2 is a plan view of the periphery of the communication device 30. FIG. 3 is a cross-sectional view taken along the line A-A of FIG. 2. In FIGS. 1 to 3, some members unnecessary for explanation are not shown. For example, in FIGS. 1 and 2, illustration of the resin cover 34 is omitted. In FIGS. 1 to 3, the rear panel 52 and/or the side member outer panel 50 are not shown. In the drawings, “Up”, “Fr” and “Rh” indicate the upper side, the front side and the right side of the vehicle, respectively. Further, in FIG. 3, black ovals indicate joining points where members are joined by welding.

(8) As shown in FIG. 1, a roof panel 10 is disposed above the vehicle. Side member outer panels 50 are connected to both left and right sides of the roof panel 10. The side member outer panel 50 is a panel material constituting a side surface of the vehicle.

(9) The roof panel 10 is a panel material constituting the upper surface of the vehicle, and is usually made of metal such as steel. The roof panel 10 includes a design surface 12 and a placement surface 14. The design surface 12 is an outer surface of the roof panel 10 and is exposed to the outside of the vehicle. The design surface 12 occupies most of the roof panel 10. The placement surface 14 is a surface slightly recessed downward from the design surface 12. The placement surface 14 is disposed at the rear end of the roof panel 10. The placement surface 14 is covered with a resin cover 34 (see FIG. 3) described later.

(10) Here, in the present specification, a portion of the roof panel 10 behind the communication opening 20 is referred to as a “rear end portion 82”. As shown in FIG. 3, the rear end portion 82 includes a front wall portion 84 extending in a substantially horizontal direction, a vertical wall portion 86 extending downward from the rear end of the front wall portion 84, and a rear wall portion 88 extending rearward from the lower end of the vertical wall portion 86. The rear end portion 82 forms a first closed space 70 and a second closed space 72 between the rear header 60 and the rear end portion 82.

(11) On the placement surface 14, a communication opening 20, a recessed portion 22, and a plurality of beads 24 are formed. The communication opening 20 is a substantially rectangular opening elongated in the vehicle width direction. For example, the dimension of the communication opening 20 in the vehicle width direction is $\frac{1}{2}$ or more of the dimension of the roof panel 10 in the vehicle width direction. The communication opening 20 is formed in order to

transmit radio waves well. The communication module **30a** and the antenna **30b** described later are disposed inside the communication opening **20** in plan view.

(12) A total of two recessed portions **22** are formed on both sides of the communication opening **20** in the vehicle width direction. The recessed portion **22** is a portion depressed further toward the vehicle inner side (i.e., the lower side) from the placement surface **14**. The recessed portion **22** is connected to an end portion of the communication opening **20** in the vehicle width direction. The recessed portion **22** is substantially rectangular in plan view.

(13) The bead **24** is a convex bead protruding from the recessed portion **22**. The bead **24** extends in the longitudinal direction of the vehicle. The top surface of the bead **24** is lower than the placement surface **14**. In other words, the bead **24** does not protrude outward from the design surface **12** of the roof panel **10**. By forming the bead **24**, the rigidity of the roof panel **10** is improved, and deformation of the roof panel **10** around the communication opening **20** is effectively suppressed.

(14) The communication module **30a** and the antenna **30b** are fixed to the roof panel **10** so as not to protrude outward from the design surface **12** of the roof panel **10**. The antenna **30b** transmits and receives radio waves for data communication. In this example, the antenna **30b** has a flat plate-shaped or box-shaped main body. The communication module **30a** is a device that transmits and receives data to and from other communication devices. The communication module **30a** controls transmission and reception of radio waves via the antenna **30b**. Hereinafter, when the communication module **30a** and the antenna **30b** are not distinguished from each other, they are referred to as a “communication device **30**”.

(15) Both the communication module **30a** and the antenna **30b** are positioned inside the communication opening **20** in a plan view. In other words, the communication module **30a** and the antenna **30b** are not covered with the roof panel **10** made of a steel material. When the resin cover **34** is detached, both the communication module **30a** and the antenna **30b** are exposed to the outside. With such a configuration, since the radio wave for communication is not blocked by the roof panel **10**, the radio wave can be satisfactorily transmitted and received.

(16) The communication module **30a** and the antenna **30b** are attached to the antenna plate **38** (see FIG. 3) by a fastening member (not shown) such as a bolt. The antenna plate **38** is a plate material that supports the communication module **30a** and the antenna **30b**. As shown in FIG. 3, the antenna plate **38** includes a mounting portion **38a** to which the communication device **30** is mounted, a hanging portion **38b** extending from the rear end of the mounting portion **38a** toward the vehicle rear side, and a terminal portion **38c** extending from the lower end of the hanging portion **38b** toward the vehicle rear side. The antenna plate **38** is fastened to a roof reinforcement **40** (hereinafter referred to as a “roof RF **40**”) and a rear header **60**, which will be described later, by a fastening member such as a bolt. In other words, the antenna plate **38** and the communication device **30** are fixed to the roof panel **10**. Fastening of the antenna plate **38** to the roof RF **40** and the rear header **60** is performed, for example, from the cabin **58** side.

(17) As shown in FIG. 3, the roof RF **40** is a panel material bonded to the lower surface of the roof panel **10** in order to reinforce the roof panel **10**. The roof RF **40** is disposed near the front end of the communication opening **20**. The roof RF **40** is fixed to the roof panel by welding or the like. The roof RF **40** has a substantially L-shaped cross-sectional shape. The roof RF **40** forms a front closed space **44** elongated in the vehicle width direction between the roof RF **40** and the roof panel **10**.

(18) The rear header **60** is also a panel material bonded to the lower surface of the roof panel **10** in order to reinforce the roof panel **10**. The rear header **60** is disposed near the rear end of the communication opening **20**. The rear header **60** is fixed to the roof panel **10** by welding or the like.

(19) The rear header **60** includes a front wall portion **62**, a vertical wall portion **64** extending from a rear end of the front wall portion **62** toward a lower side, and a rear wall portion **66** extending from a lower end of the vertical wall portion **64** toward a rear side. Further, a groove-shaped portion **68** extending in the vehicle width direction is formed in the rear wall portion **66**. The groove-shaped

portion **68** forms a first closed space **70** elongated in the vehicle width direction between the groove-shaped portion **68** and the rear wall portion **88** of the roof panel **10**. Further, a slight unevenness is formed on the vertical wall portion **64**. Due to the irregularities, the vertical wall portion **64** forms a second closed space **72** elongated in the vehicle width direction between the vertical wall portion **64** and the vertical wall portion **86** of the roof panel **10**.

(20) The rear header **60** is joined to the roof panel **10** at a first joining point **74** located at a rear side of located at a rear side of the vehicle relative to the groove-shaped portion **68**, a second joining point **76** located at the vertical wall portion **64**, and a third joining point **78** located at the front wall portion **62**. Here, the first joining point **74** and the third joining point **78** can be regarded as the rear end and the front end of the rear header **60**. Accordingly, the rear header **60** is joined to the roof panel **10** at the front end and the rear end thereof. The terminal portion **38c** of the antenna plate **38** is connected to the rear wall portion **66** of the rear header **60**. The reason why the rear header **60** has such a configuration will be described later.

(21) As shown in FIG. 3, the resin cover **34** covers the entire placement surface **14** from above. In other words, the resin cover **34** covers the entire bead **24** and the communication device **30** from above. Therefore, neither the bead **24** nor the communication device **30** is visible from outside the vehicle. The protruding amount of the bead **24** is such that the bead **24** does not contact the resin cover **34**.

(22) The resin cover **34** is made of a resin that transmits radio waves, for example, a general-purpose resin such as polypropylene or polycarbonate/polyester. The resin cover **34** is fixed to the roof panel **10** by, for example, adhesion. Needless to say, the resin cover **34** may be fixed to the roof panel **10** by a fastening member such as a bolt or a clip. In this case, the sealing member may be disposed on the outer peripheral portion of the resin cover **34**. The upper surface of the resin cover **34** is smoothly continuous with the design surface **12** of the roof panel **10**.

(23) A roof head lining **54** is disposed below the antenna plate **38**. The roof head lining **54** is a design panel functioning as a ceiling of the cabin **58**. In the vicinity of the rear end of the cabin **58**, the roof head lining **54** gradually curves downward so as not to interfere with the antenna plate **38** in the course of extending toward the rear side of the vehicle. The roof head lining **54** is bent largely at the bending point **56** and then extends toward rear side of the vehicle.

(24) As apparent from the above description, in the vehicle upper structure disclosed in the present specification, the front end and the rear end of the rear header **60** are joined to the roof panel **10**, and the closed spaces **70** and **72** are formed between the rear header **60** and the roof panel **10**. The reason for this configuration will be described in comparison with the reference example. FIG. 4 is a cross-sectional view showing an upper structure of a vehicle according to a reference example.

(25) Recently, an on-vehicle communication device **30** (e.g., an antenna **30b**) has increased in size for the purpose of improving communication performance. As a result, the size of the communication opening **20** for transmitting radio waves is also increased. Here, when the large communication opening **20** is formed in the roof panel **10**, it is needless to say that the rigidity of the whole vehicle body including the roof panel **10** decreases around the communication opening **20**. As a result, deformation of the vehicle body may occur around the communication opening **20**.

(26) Therefore, in order to improve the rigidity of the vehicle body, it is conceivable to form a skeleton-shaped portion around the communication opening **20**. For example, in the reference example shown in FIG. 4, the front closed space **44** is formed near the front end of the communication opening **20**, and the rear closed space **94** is formed near the rear end of the communication opening **20**. The closed spaces **44** and **94** function as a skeleton-shaped portion. By forming the closed spaces **44** and **94**, the section modulus is increased, and the rigidity of the vehicle body around the communication opening **20** is improved.

(27) The vehicle upper structure of the reference example shown in FIG. 4 includes a rear header lower **90** and a rear header upper **92**. The rear header upper **92** has a hat-shaped cross-sectional shape opening downward. The rear header upper **92** is disposed above the rear header lower **90**.

The front end and the rear end of the rear header upper **92** are joined to the rear header lower **90**. Thereby, a rear closed space **94** elongated in the vehicle width direction is formed between the rear header upper **92** and the rear header lower **90**. With this configuration, the rigidity of the vehicle body in the vicinity of the rear end of the communication opening **20** can be improved.

(28) However, in this case, it is necessary to arrange the rear closed space **94** on the rear side of the communication device **30**. Therefore, the placement position of the communication device **30** is shifted toward the front side of the vehicle by the dimension of the rear closed space **94**. When the placement position of the communication device **30** deviates toward the front side of the vehicle, the bending point **56*** of the roof head lining **54*** also deviates toward the front side of the vehicle. Thus, the space of the cabin **58** is narrowed. In particular, the distance between the head of the occupant sitting on the rear seat and the roof head lining **54*** (hereinafter referred to as “head clearance”) becomes small, and there is a risk of giving a sense of compression to the occupant.

(29) On the other hand, in the case of the vehicle upper structure of this example, the rear closed space **94** does not exist on the rear side of the communication device **30**. As a result, the communication device **30** can be disposed on the rear side of the vehicle as compared with the reference example. Thereby, the bending point **56** of the roof head lining **54** can also be disposed on the rear side of the vehicle as compared with the reference example. As a result, the space of the cabin **58**, in particular, the head clearance can be widened.

(30) FIG. 5 is a diagram illustrating a difference in head clearance depending on the position of the bending point **56**. In FIG. 5, a solid line indicates the roof head lining **54** employed in the vehicle upper structure of the present example, and a two-dotted line indicates the roof head lining **54*** employed in the vehicle upper structure of the reference example.

(31) As is apparent from FIG. 5, the bending point **56** in the present example is positioned on the rear side of the vehicle in comparison with the bending point **56*** in the reference example. Therefore, the head clearance C_a in this example is significantly larger than the head clearance C_b in the reference example. As a result, according to the vehicle upper structure of the present example, the pressure feeling of the occupant **100** can be greatly reduced.

(32) Incidentally, when the rear closed space **94** is eliminated, the rigidity of the vehicle body in the vicinity of the rear end of the communication opening **20** decreases by a corresponding amount. In this example, in order to prevent such a decrease in rigidity, the rear header **60** is joined to the rear end portion **82** of the roof panel **10**. More specifically, as shown in FIG. 3, the rear header **60** is joined to the roof panel **10** at the rear end (i.e., the first joining point **74**) and the front end (i.e., the second joining point **76**) thereof. Thereby, the first closed space **70** is formed between the groove-shaped portion **68** of the rear header **60** and the rear wall portion **88** of the roof panel **10**. Since the first closed space **70** functions as a skeleton-shaped portion, the rigidity of the vehicle body in the vicinity of the rear end of the communication opening **20** is improved by forming the first closed space **70**.

(33) The rear header **60** is also joined to the roof panel **10** at the second joining point **76** located at the vertical wall portion **64**. By providing the second joining point **76** in the vertical wall portion **86** extending in the vertical direction, the rigidity of the vehicle body in the vicinity of the rear end of the communication opening **20** can be improved while suppressing the dimension in the vehicle longitudinal direction. As a result, the space of the cabin **58** and the head clearance can be sufficiently ensured while ensuring the rigidity.

(34) Further, a second closed space **72** is formed between the second joining point **76** and the third joining point **78**. With this configuration, the second closed space **72** functions as a second skeleton-shaped portion. Thus, the rigidity of the vehicle body in the vicinity of the rear end of the communication opening **20** can be further improved.

(35) Here, if the torsional rigidity of the vehicle body is low, the match box deformation tends to occur. The match box deformation is a phenomenon in which the entire vehicle body is deformed like a match box which is collapsed obliquely. In order to suppress such a match box deformation,

it is required that the rigidity of the vehicle body is high at an end portion in the vehicle width direction, that is, at a so-called corner portion. Therefore, in this example, the density of the joining points **74**, **76**, **78** at the end region in the vehicle width direction Re (see FIG. **1**) is higher than the density of the joining points **74**, **76**, **78** at the center region in the vehicle width direction Rc . For example, a plurality of first joining points **74** are set at intervals in the vehicle width direction. The interval between the plurality of first joining points **74** in the vehicle width direction may be changed according to the position in the vehicle width direction. For example, the interval at end region in the vehicle width direction Re may be narrower than the interval at center region in the vehicle width direction Rc . The same applies to the second joining point **76** and the third joining point **78**. Further, at the end region in the vehicle width direction Re , a plurality of joining points **74**, **76**, **78** may be provided at intervals in the vertical direction or the front-rear direction. For example, at the end region in the vehicle width direction Re , a plurality of second joining points **76** may be provided at intervals in the vertical direction. By providing many joining points **74**, **76**, **78** at the end portion in the vehicle width direction, the rigidity at the end portion in the vehicle width direction can be improved. As a result, the match box deformation can be effectively suppressed.

(36) In the vehicle upper structure of this example, the rear portion of the antenna plate **38** is formed in a substantially L-shape having a hanging portion **38b** and a terminal portion **38c**. With such a configuration, the section modulus of the antenna plate **38** can be increased while effectively utilizing the space. That is, around the communication device **30**, both the roof head lining **54** and the roof panel **10** extend downward as they approach the rear end of the vehicle. Therefore, there is no spatial margin above the rear portion of the antenna plate **38**. On the other hand, there is a space margin under the rear portion of the antenna plate **38**. In this example, since the rear portion of the antenna plate **38** extends toward the lower side of the rear portion of the antenna plate **38** having the spatial margin, the vertical dimension of the antenna plate **38** can be increased without narrowing the cabin **58**. As a result, the space can be effectively utilized while the rigidity of the vehicle body around the communication opening **20** is kept high.

(37) Any of the configurations described above is an example. Other configurations may be varied as long as the rear header **60** is joined to the roof panel **10** at its front and rear ends and a first closed space **70** is formed between the rear header **60** and the roof panel **10**. For example, the second joining point **76** and the second closed space **72** may be eliminated. In the above description, the recessed portions **22** and the beads **24** are disposed on both sides of the communication opening **20** in the vehicle width direction. Further, the shape of each member may be appropriately changed.

REFERENCE SIGNS LIST

(38) **10** roof panel, **12** design surface, **14** placement surface, **20** communication opening, **22** recessed portion, **24** bead, **30** communication device, **30a** communication module, **30b** antenna, **34** resin cover, **38** antenna plate, **38a** mounting portion, **38b** hanging portion, **38c** terminal portion, **40** roof reinforcement, **44** closed space, **50** side member outer panel, **52** rear panel, **54** roof head lining, **56** bending point, **58** cabin, **60** rear header, **62** front wall portion, **64** vertical wall portion, **66** rear wall portion, **68** groove-shaped portion, **70** closed spaces, **72** second closed space, **74** first joining point, **76** second joining point, **78** third joining point, **82** rear end portion, **84** front wall portion, **86** vertical wall portion, **88** rear wall portion, **90** rear header lower, **92** rear header upper, **94** rear closed space, **100** occupant.

Claims

1. A vehicle upper structure comprising: a roof panel having a communication opening formed therein; a roof reinforcement disposed near a front end of the communication opening and joined to the roof panel; a rear header disposed near a rear end of the communication opening and joined to the roof panel; and an antenna plate that supports a communication device and is connected to the

roof reinforcement and the rear header, wherein the rear header is joined to the roof panel at a front end and a rear end thereof, and forms a first closed space elongated in a vehicle width direction between the rear header and the roof panel, wherein the rear header comprising: a front wall portion; a vertical wall portion extending from a rear end of the front wall portion toward a lower side of the vehicle; a rear wall portion extending from a lower end of the vertical wall portion toward a rear side of the vehicle; and a groove-shaped portion formed in the rear wall portion and forming the first closed space between the groove-shaped portion and the roof panel, and wherein the rear header is joined to the roof panel at a first joining point located at a rear side of the vehicle relative to the groove-shaped portion, a second joining point located at the vertical wall portion, and a third joining point located at the front wall portion.

2. The vehicle upper structure according to claim 1, wherein the rear header forms a second closed space between the rear header and the roof panel at a portion between the second joining point and the third joining point.

3. The vehicle upper structure according to claim 1, wherein: the antenna plate includes a mounting portion to which the communication device is mounted, a hanging portion extending from a rear end of the mounting portion toward a lower side of the vehicle, and a terminal portion extending from a lower end of the hanging portion toward a rear side of the vehicle; and a terminal portion of the antenna plate is connected to the rear wall portion of the rear header.

4. A vehicle upper structure comprising: a roof panel having a communication opening formed therein; a roof reinforcement disposed near a front end of the communication opening and joined to the roof panel; a rear header disposed near a rear end of the communication opening and joined to the roof panel; and an antenna plate that supports a communication device and is connected to the roof reinforcement and the rear header, wherein the rear header is joined to the roof panel at a front end and a rear end thereof, and forms a first closed space elongated in a vehicle width direction between the rear header and the roof panel, wherein a density of a joining point between the rear header and the roof panel at an end region in the vehicle width direction of the roof panel is higher than the density of the joining point at a center region in the vehicle width direction of the roof panel.
